STAT2004/2904/7004 – Assignment 3

Due: 09 October 2026 @ 4pm

STAT2004/7004: *Complete Exercises 1–4 for a maximum of 40 marks and a total of 10%.*

STAT2904: *Complete Exercises 1–5 for a maximum of 50 marks and a total of 10%.*

All students: *Exercise 6 is a bonus question. A complete and correct solution of this question earns you an extra 1% for this assignment.*

Note that some questions involve ***interpretation and communication of results in the form of an audio recording*** *which you upload onto Blackboard as an audio file.*

Question 1. **[7 marks]**

Measurements from a physics lab machine follow a normal distribution with some mean μ and some variance σ^2. To investigate the precision of this machine, 18 independent measurements, $X_1, X_2, \ldots, X_{18}$, are taken from this machine at random times over a period of three days. Based on the value of the sample variance

$$S^2 = \frac{1}{17}\sum_{i=1}^{18}(X_i - \bar{X})^2,$$

two competing confidence intervals for the unknown variance σ^2 are proposed:

$$\begin{aligned} &\text{Ross's interval:} && (0.563\,S^2,\ 2.247\,S^2)\ , \\ &\text{Alan's interval:} && (0.598\,S^2,\ 2.653\,S^2)\ . \end{aligned}$$

(a) (*1 mark*) State the distribution of the transformed random variable, $17\,S^2/\sigma^2$.

(b) (*4 marks*) Using your result from part (a), or otherwise, show that both intervals will contain the unknown σ^2 with 95% confidence.

(c) (*2 marks*) Which of these two 95% confidence intervals is "better"? Briefly justify your choice.

Question 2 (Prior and posterior beliefs) [11 marks]

Let's re-consider the Skittles versus M&Ms example from the lectures. Suppose that our prior belief about the underlying true proportion p of Skittles is characterized by

$$f(p) \ \propto\ (1-p)^7\ , \quad 0 \le p \le 1\ .$$

(a) *(1 mark)* Graph this prior belief $f(p)$ as a function of p across the range $0 \le p \le 1$.

(b) *(1 mark)* Qualitatively describe our prior belief. In particular, around what values of p is our prior belief the strongest, and around what values of p is our prior belief the the weakest?

(c) *(2 marks)* After carrying out our experiment, we observed $x = 23$ Skittles out of $n = 47$ samples. Show how our prior belief should be updated after observing these data by computing the posterior belief.

(d) *(1 mark)* Plot the posterior belief from part (c) over the range $0 \le p \le 1$.

(e) *(1 mark)* Qualitatively describe how our belief about p has changed after observing the data.

(f) *(3 mark)* Construct and interpret a 95% credible interval for the underlying true proportion p based on our posterior belief.

(g) *(2 marks)* By comparing all priors and posteriors that we covered in class, tutorials, and this assignment question, comment on:

(i) the effect of the prior belief on the posterior belief ;
(ii) the effect of the data on the posterior belief.

Question 3 (Confidence interval for a bound parameter) [11 marks]

Let $X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} \text{Uniform}(-\theta, \theta)$, where $\theta \geq 0$ is an unknown parameter.

(a) (*2 marks*) State the distribution of $U_1 = |X_1|/\theta$.

(b) (*2 marks*) Show that the MLE of θ is given by $\hat{\theta} = \max\{|X_1|, |X_2|, \ldots, |X_n|\}$.

(c) (*2 marks*) Using part (a), or otherwise, show that $U_{(n)} = \hat{\theta}/\theta$ has cumulative distribution given by

$$P\big(U_{(n)} \leq u\big) = \begin{cases} 0\,, & u < 0 \\ u^n\,, & 0 \leq u \leq 1 \\ 1\,, & u > 1 \end{cases}$$

(d) (*1 mark*) Is $U_{(n)}$ a pivot quantity for θ? Briefly explain your answer.

(e) (*1 mark*) Using part (c), or otherwise, find a pair of critical values a and b such that $P(a \leq U_{(n)} \leq b) = 95\%$.

(f) (*2 marks*) A sample of $n = 7$ observations is taken from this distribution, giving the following data values:

1.421 –1.197 0.476 1.106 –2.456 1.865 –0.581

Using parts (d) and (e), or otherwise, construct and interpret a 95% confidence interval for θ based on these data.

(g) (*1 mark*) [**Audio question**]: A STAT2004 student concludes that "*there is a 95% chance that the true θ is contained within the interval calculated in part (f)*". Do you agree with this conclusion? Briefly explain why, or why not.

Question 4 (Hypothesis testing for a failure probability) [11 marks]

Suppose that a current industry-standard engineering model predicts that a certain type of concrete block will have a 40% chance of failure when subject to a two-tonne load. A civil engineer wants to test the accuracy of this claim by obtaining 50 independent samples of such blocks, subjecting each to a two-tonne load and then recording the total number X of blocks that broke. She decides that she will reject the claim $H_0 : p = 0.4$ in favour of $H_1 : p \neq 0.4$ if $X \leq 15$ or $X \geq 26$.

(a) (*1 mark*) What is the significance level α of this test?

(b) (*2 marks*) Graph the power of this test as a function of p.

The civil engineer proposes an alternative model for predicting the chance of failure of concrete blocks. Under the alternative model, the probability of failure for this type of concrete block is predicted to be 50%. She wants to test whether her new model is more accurate than the current industry standard.

(c) (*3 marks*) Construct a likelihood ratio test (LRT) of $H_0 : p = 0.4$ versus $H_1 : p = 0.5$ based on the number of failures X from 50 trials.

(d) (*1 mark*) Show that the LRT is equivalent to the test with rejection region $\{X \geq c\}$ for some cutoff c.

(e) (*1 mark*) Changes in industry standards should not be taken lightly, so a significance level of 1% is often used in engineering instead of 5%. Find the cutoff value c for carrying out this test at the 1% significance level.

(f) (*1 mark*) If the alternative model is actually correct, what is the power of this test?

(g) (*2 marks*) What sample size is needed to increase the power of this test to 90%, while keeping the level of the test at 1%?

Question 5 (STAT2904 only) [10 marks]

Let $X_1, X_2, \ldots, X_n$ be an iid sample from a continuous distribution with pdf

$$f_\theta(x) = \frac{2x}{\theta^2} \ , \qquad 0 \leq x \leq \theta \ ,$$

where $\theta > 0$ is an unknown parameter.

(a) (*2 marks*) Derive the MLE $\hat{\theta}$ of θ.

(b) (*2 marks*) Find the cdf of $\hat{\theta}$.

(c) (*1 mark*) Using part (b), construct a pivot for the parameter θ via the cumulative probability transform method. State the distribution of your pivot variable.

(d) (*2 marks*) Using your pivot from part (c), show that *any* interval of the form,

$$\left(\frac{X_{(n)}}{(a+0.95)^{1/2n}}, \frac{X_{(n)}}{a^{1/2n}} \right) \ , \quad 0 \leq a \leq 0.05 \ ,$$

defines a 95% confidence interval for θ.

(e) (*2 marks*) Find the value a that minimises the length of the interval from part (d).

(f) (*1 mark*) Enumerate the shortest 95% confidence interval for θ based on the following realisations from $f_\theta(x)$: 1.227 3.829 4.813 4.874 5.142.

Question 6 (Bonus Question for all students) [4 marks]

Let $X_1, X_2, \ldots, X_{17}$ be a random sample of size $n = 17$ coming from some arbitrary continuous distribution with unknown population median, m.

(a) Show that the interval,

$$\left[X_{(5)}, X_{(13)} \right] \ ,$$

defines an (approximate) 95% confidence interval for the population median, m. Here, $X_{(5)}$ denotes the fifth smallest observation, and $X_{(13)}$ denotes the fifth largest observation.

Confidence intervals such as these for the population median are intriguing because they depend only on order statistics and do not require any actual model for the underlying distribution of the data!

(b) Suppose the underlying population distribution is actually t_2, which has a median (and mean) equal to 0. Using computer simulations, or otherwise, compare the relative merits of the interval from part (a) and the classical $\bar{X} \pm t_{16}^{(0.975)} S/\sqrt{17}$ confidence interval for capturing the true mean/median of 0.

You may use the following `R` code to simulate $n = 17$ observations from a t_2 distribution:

```
> rt(17, 2)
```